

The Effect of Federal Reserve Rates on Net Interest Margins: 2016 to 2023

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Abstract

This study examines the relationship between Federal Reserve interest rates and net interest margins (NIM) among major U.S. banks from 2016 to 2023. Data were collected for JPMorgan Chase, Bank of America, and Wells Fargo, alongside Federal Funds Rate data from the Federal Reserve Economic Data (FRED) database. A quantitative approach using scatterplots and linear regression analysis was employed to evaluate the relationship between Federal Reserve interest rate changes and bank net interest margins. The regression analysis results indicate a moderate, positive relationship between Federal Reserve interest rate changes and bank net interest margins with slight variations between the financial institutions. These findings imply that while the Federal Reserve interest rate changes are associated with net interest margins, additional bank-specific and macroeconomic factors also played a role.

Keywords: Federal Reserve interest rates, net interest margins, regression

The Effect of Federal Reserve Rates on Net Interest Margins: 2016 to 2023

The Federal Reserve plays a significant role in the United States economy through the adjustment of interest rates used to influence inflation, employment, and economic growth. Changes in the Federal Funds Rate directly affect borrowing costs across the financial system, influencing consumer loans, mortgages, and commercial lending activity. Because banks generate a significant portion of their revenue through lending, changes in Federal Reserve interest rates can strongly influence bank profitability.

One of the most important indicators of bank profitability is net interest margin (NIM), which measures the difference between interest income earned on assets and interest paid on liabilities relative to earning assets. Higher interest rates often allow banks to increase loan yields, although the extent of this benefit depends on factors such as deposit costs, balance sheet composition, and business structure.

This study examines the relationship between Federal Reserve interest rates and net interest margins from 2016 to 2023 among three major U.S. banks: JPMorgan Chase, Bank of America, and Wells Fargo. This time period includes both the near-zero interest rate environment during the COVID-19 pandemic and the aggressive rate increases that followed, making it particularly useful for analyzing how changing monetary policy influences bank profitability.

Method

Obtaining and Measuring Data

To obtain the net interest margins of JPMorgan Chase, Bank of America, and Wells Fargo, I searched their respective 10-K's for each year. This step required adaptability as each bank's net interest margin was termed differently across each bank

and was in a different location for each bank. To obtain the Federal Reserve interest rates for each year, I used the FRED database.

Assessments and Measures

The independent variable was Federal Reserve interest rates, representing U.S. monetary policy conditions. The dependent variable was the net interest margins (NIM), measured as net interest income relative to average earning assets. Linear regression and the correlation coefficient (R^2) were used to evaluate the relationship between the independent and dependent variables.

Results

A thorough financial analysis was conducted to evaluate the relationship between Federal Reserve interest rates and net interest margins (NIM) for JPMorgan Chase, Bank of America, and Wells Fargo from 2016 to 2023.

Bank	Regression Equation (x=Fed funds rate)	R ² Value	Relationship Strength
JPMorgan Chase	$0.162 \cdot x + 1.98$	0.568	Moderate Positive
Bank of America	$0.149 \cdot x + 1.75$	0.489	Moderate Positive
Wells Fargo	$0.129 \cdot x + 2.43$	0.33	Weak Positive

Figure 1 Bank Regression Equation, R² Value, and Relationship Strength

Across all three banks, the results indicate a moderate positive relationship between Federal Reserve interest rates and net interest margins, meaning that higher interest rates are generally associated with higher bank profit margins through increased net interest margins. However, the exact strength of this relationship varied across institutions.

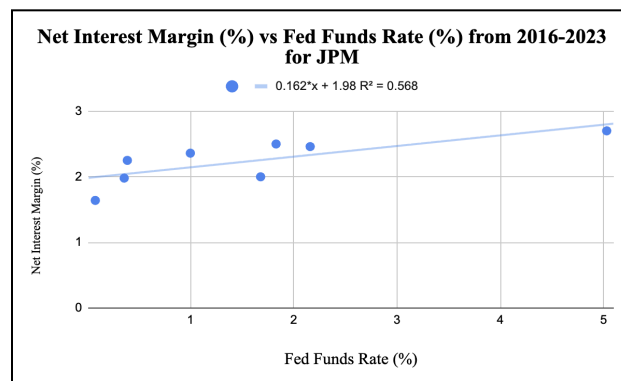


Figure 2 Net Interest Margin (%) vs Fed Funds Rate (%) for JPMorgan Chase

For JPMorgan Chase, the regression analysis produced a moderate, positive relationship, with net interest margins showing a relatively consistent response to changes in interest rates over time. This suggests that JPMorgan Chase's earning structure is more directly sensitive to monetary policy changes.

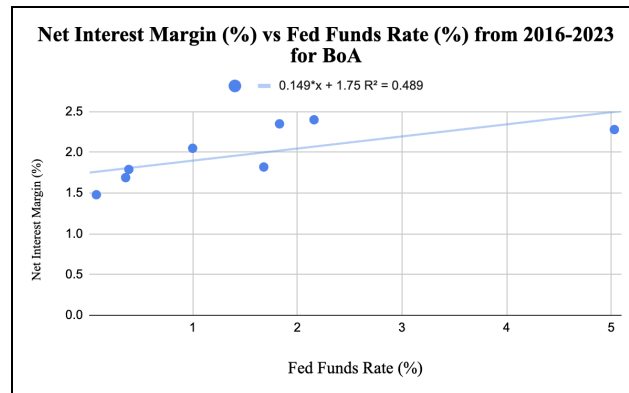


Figure 3 *Net Interest Margin (%) vs Fed Funds Rate (%) for Bank of America*

For Bank of America, the relationship was also positive but showed greater variability. While net interest margins increased during periods of rising interest rates, the data were less stable, indicating additional influencing factors beyond Federal Reserve policy.

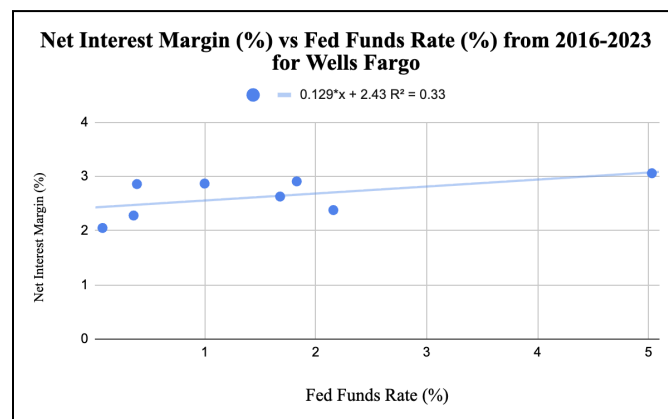


Figure 4 *Net Interest Margin (%) vs Fed Funds Rate (%) for Wells Fargo*

For Wells Fargo, the relationship appeared much weaker than the other two institutions, and showed less variation during the COVID-19 low-rate period compared to the other banks. This suggests a difference in balance sheet composition and interest rate exposure.

Overall, regression results showed varying slopes and R^2 values across banks, indicating that while Federal Reserve interest rates are an important factor influencing net interest margins, they do not fully explain changes in bank profitability over time.

Limitations

The results of this study should be considered with some caution. Firstly, the sample size consisted of only 8 annual observations from 2016 to 2023. A larger dataset with quarterly or monthly observations may provide stronger statistical reliability. Additionally, the analysis focused solely on the Federal Funds Rate as an explanatory variable, despite the fact that net interest margins are influenced by many other factors such as yield curve shape, loan demand, deposit costs, and macroeconomic conditions. Finally, the use of regression analysis identified association rather than causation, meaning that Federal Reserve interest rates can not be concluded to be the sole cause of changes in bank profitability.

Discussion

The findings of this study suggest that there is a positive, while moderate, correlation between Federal Reserve interest rates and net interest margins of major U.S. banks, although the strength of this correlation varies across institutions. This supports the idea that monetary policy transmission into bank profitability is heterogeneous, depending on each bank's structure and business model.

From a financial perspective, this relationship exists because higher interest rates generally allow banks to earn more from lending activities compared to what they pay on deposits. However, the differences observed between the different banks highlight important structural factors. For example, banks with larger proportions of retail deposits and diversified revenue streams may be more sensitive to changes in interest rates as opposed to banks more reliant on traditional lending.

The analytical process used in this study, the combination of tables, line graphs, and scatterplots, allowed for both visual and statistical interpretation of these relationships. The slope

of each regression line represented the sensitivity of net interest margin to interest rate changes, while the R^2 value indicated how much of the variation in net interest margin could be explained by Federal Reserve policy. The moderate R^2 values observed suggest that interest rates are a meaningful but incomplete predictor of bank profitability.

These results have broader implications for understanding the banking sector and monetary policy. For investors, the findings suggest that interest rate environments can significantly impact bank profitability, but bank selection matters due to differing sensitivities. For policymakers, the results indicate that changes in the Federal Funds Rate may have uneven effects across financial institutions, which could influence credit availability and financial stability in different sectors of the economy.

Beyond this study, future analysis could incorporate additional variables such as yield curve shape, credit risk, and loan growth to better capture the complexity of bank profitability. Expanding the model beyond a single explanatory variable would likely improve explanatory power and provide a more complete understanding of how macroeconomic conditions influence financial institutions.

Conclusion

This study found a moderate positive relationship between Federal Reserve interest rates and the net interest margins (NIM) from 2016 to 2023 of three major banks in the U.S.: JPMorgan Chase, Bank of America, and Wells Fargo. While all three banks demonstrated an upward trend, they all had different strengths, representing the fact that each bank's individual policies and business compositions may have influenced the sensitivity of the institutions to Federal Reserve interest rates on their individual net interest margins (NIM). Although Federal Reserve.

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Appendix

Net Interest Margin (%) of each Major U.S. Bank and Fed Funds Rate (%) from 2016 to 2023

	2016	2017	2018	2019	2020	2021	2022	2023
Net Interest Margin (%) for JPMorgan Chase	2.25	2.36	2.5	2.46	1.98	1.64	2	2.7
Net Interest Margin (%) for Bank of America	1.79	2.05	2.35	2.4	1.69	1.48	1.82	2.28
Net Interest Margin (%) for Wells Fargo	2.86	2.87	2.91	2.38	2.28	2.05	2.63	3.06
Fed Funds Rate (%)	0.39	1	1.83	2.16	0.36	0.08	1.68	5.03

